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Riesz Transform Associated with Linear Canonical Transform: Definition and Its Application in Envelope Detection

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ABSTRACT

This article presents a novel approach called the generalized Riesz transform (GRT) that addresses the issue of negative frequency components in linear canonical transform (LCT). The GRT is an isotropic extension of the two-dimensional Hilbert transform and finds applications in computer vision, medical imaging, and pattern recognition, with specific emphasis on edge detection, texture analysis, and feature extraction. The article defines the GRT within the framework of linear canonical domains using three-dimensional harmonic fields and properties of the LCT, and provides a spatial domain filtering formula for GRT. The effectiveness of GRT in capturing texture features is demonstrated through the concept of generalized monogenic signal, and an envelope detection algorithm based on GRT and phase consistency is proposed, showcasing superior performance compared to conventional methods.

CCS CONCEPTS

• **Applied computing** → **Mathematics and statistics**; *Electronics*;
• **Theory of computation** → *Theory and algorithms for application domains*.

KEYWORDS

Generalized Riesz transform, linear canonical transform, generalized monogenic signal, envelope detection

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1 INTRODUCTION

The Riesz transform (RT) is an isotropic two-dimensional (2D) extension of the Hilbert transform (HT) [4, 8, 24]. Prior to the introduction of the Riesz transform, several scholars had made notable attempts to extend the Hilbert transform to high-dimensional signal spaces. For instance, the partial Hilbert transform (PHT) and the total Hilbert transform (THT) were proposed as extensions [2, 7, 13].

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In comparison to these methods, the feature extraction process of the RT possesses key attributes such as rotation invariance and scale invariance. These properties render the RT particularly suitable for tasks that heavily rely on the orientation and scale of features in the data [10, 20].

Several studies have underscored the significance of the RT across diverse fields, including computer vision, motion estimation, and medical analysis [11, 14, 23]. These studies serve as compelling evidence for the broad range of applications of the RT, encompassing tasks like edge detection, texture analysis, and feature extraction in signal and image processing [3, 12, 17]. Moreover, these studies effectively demonstrate how the analysis of RT coefficients can yield valuable insights into local orientation, frequency content, and energy distribution within the data. Consequently, such insights contribute significantly to the overall comprehension and utilization of the RT in various domains.

The linear canonical transform (LCT)[1, 9, 18] is a three-free-parameter (a, b, c, d) class of integral transform. It is a generalization of the FT, the fractional Fourier transform (FRFT), and the Fresnel transform (FST), making it play a more significant role in signal processing. Given the significant role of LCT in signal analysis, it is essential to study the 2D Hilbert transform as a means to suppress its negative frequency components. Currently, research has expanded the applications of the PHT and the THT to linear canonical domains [13]. However, these transformations still exhibit anisotropy. Therefore, the objective of this article is to develop an isotropic generalized Riesz transform (GRT) utilizing the LCT.

This article initially establishes the definition of the GRT in the linear canonical domain by utilizing the three-dimensional (3D) harmonic field and the fundamental properties of LCT. It further presents the filtering formulation of GRT in the spatial domain. Subsequently, the application of GRT in image envelope detection is discussed, highlighting its effectiveness in capturing texture features through the definition of generalized monogenic signal (GMS). Additionally, an envelope detection algorithm based on GRT and phase consistency [19, 22] is proposed and compared with conventional envelope detection algorithms, demonstrating its superior performance.

2 PRELIMINARIES

2.1 Riesz Transform

The Riesz kernels in $\mathbf{t} = (t_1, t_2)$ and their 2D (with 2 variables) Fourier transforms (FTs) are given by [16]

$$r_k(\mathbf{t}) = -\frac{t_k}{2\pi \left(\sqrt{t_1^2 + t_2^2} \right)^3} \xrightarrow{2D FT} \mathcal{R}_{k,k} = \frac{i\omega_k}{\sqrt{\omega_1^2 + \omega_2^2}} \quad (1)$$

where i represents the imaginary part and $\mathcal{H}_{R_{i,k}}$ represents the 2D FT of r , with $k = 1, 2$ as the index. Then the Riesz transforms (RTs) defined as

$$f_{R_{i,i}}(\mathbf{t}) = f(\mathbf{t}) * r_{i,i}(\mathbf{t}). \quad (2)$$

where $*$ denotes 2D convolution.

The HT is capable of constructing analytic signal representations of one-dimensional signals. Similarly, the RT can generate analytic signal representations of 2D signals, known as monogenic signal [4, 7]. However, complex algebra alone is inadequate to embed generalized 2D analytical signals, as both the signal and its Riesz transform consist of three components. The mathematical framework that deals with three-dimensional (3D) vectors is known as Clifford analysis [11]. The monogenic signal have the form [4]

$$\varphi_M(\mathbf{t}) = f - i f_{R_{1,1}}(\mathbf{t}) - i f_{R_{1,2}}(\mathbf{t}). \quad (3)$$

2.2 Linear Canonical Transform

The 2D LCT is defined by [15]

$$L_A\{f\}(\omega) = \int_{\mathbb{R}^2} f(\mathbf{t}) K_{A_1}(\mathbf{t}_1, \omega_1) K_{A_2}(\mathbf{t}_2, \omega_2) d\mathbf{t} \quad (4)$$

where $\mathbf{t} = (t_1, t_2)$, $\omega = (\omega_1, \omega_2)$, $\mathbf{A} = (A_1, A_2)$, $A_k = (a_k, b_k; c_k, d_k)$, and $K_{A_k}(t_k, \omega_k) = \frac{1}{\sqrt{i2\pi b_k}} e^{i\left(\frac{a_k}{2b_k} t_k^2 - \frac{1}{b_k} t_k \omega_k + \frac{d_k}{2b_k} \omega_k^2\right)}$, $b_k \neq 0$, $a_k d_k - b_k c_k = 1$, $k = 1, 2$. Furthermore, we define the quaternion LCT (QLCT), as mentioned earlier, which requires the use of Clifford analysis. For the sake of simplicity in the formula, let us denote $\mathbf{I} = (i, j)$, representing a vector composed of two imaginary units i and j . The QLCT of a signal $f \in L^1(\mathbb{R}^2; \mathbb{H})$ is given by [25]

$$L_{q,A}\{f\}(\omega) = \int_{\mathbb{R}^2} K_{i,A_1}(t_1, \omega_1) f(\mathbf{t}) K_{j,A_2}(t_2, \omega_2) d\mathbf{t} \quad (5)$$

where

$$K_{i,A_k}(t_k, \omega_k) = \frac{1}{\sqrt{i_k 2\pi b_k}} e^{i_k \left(\frac{a_k}{2b_k} t_k^2 - \frac{1}{b_k} t_k \omega_k + \frac{d_k}{2b_k} \omega_k^2 \right)}. \quad (6)$$

2.3 Phase consistency

Phase consistency is a metric that gauges the similarity of phase between different frequency components across different locations in an image. It is a dimensionless value that remains unaffected by variations in lighting and brightness [19, 22]. The phase consistency function is defined as

$$PC(x) = \max \frac{\sum_n A_n \cos(\phi_n(x) - \bar{\phi}(x))}{\sum_n A_n} \quad (7)$$

where A_n represents the amplitude of the n th Fourier component, ϕ_n represents the phase angle of the n th Fourier component, and $\bar{\phi}$ is the amplitude-weighted average local phase angle of all Fourier terms at the specific point that maximizes the equation.

Phase consistency is a challenging measure to compute. As an alternative, Venkatesh and Owens proposed in [22] that the maximum phase consistency point can be equivalently determined by identifying the peak in the local energy function. They argued that the local energy function is directly proportional to the phase consistency function, hence the peak in local energy corresponds to the peak in phase consistency, that is

$$E(x) = \sqrt{I^2(x) + H^2(x)} = PC(x) \sum_n A_n \quad (8)$$

where $I(x)$ represents the real component, $H(x)$ is the result of Hilbert transform from the original signal.

3 GENERALIZED RIESZ TRANSFORM

3.1 Definition derivation

For a harmonic function $u(x, y)$, an analytic function $\varphi(z) = u(x, y) + iv(x, y)$ can be constructed, where $v(x, y)$ is the harmonic conjugate function of u , meaning v and u satisfy a HT pair [24]. Therefore, the HT can be derived from a 2D harmonic field. In a similar manner, the derivation of the generalized Riesz transform (GRT) requires starting from a 3D harmonic field.

A 3D harmonic function $u(\mathbf{t})$ is a solution of Laplace equation

$$\nabla^2 u(\mathbf{t}) = \frac{\partial^2}{\partial t_1^2} u(\mathbf{t}) + \frac{\partial^2}{\partial t_2^2} u(\mathbf{t}) + \frac{\partial^2}{\partial t_3^2} u(\mathbf{t}) = 0 \quad (9)$$

where $\mathbf{t} = (t_1, t_2, t_3)^\top$, $\nabla = (\frac{\partial}{\partial t_1}, \frac{\partial}{\partial t_2}, \frac{\partial}{\partial t_3})^\top$. Denoting the harmonic potential field of $u(\mathbf{t})$ that is modulated by a complex exponential function in the t_1 and t_2 directions

$$\mathbf{v}(\mathbf{t}) = (v_1(\mathbf{t}), v_2(\mathbf{t}), v_3(\mathbf{t}))^\top = e^{-i\frac{a_1}{2b_1} t_1^2} e^{-i\frac{a_2}{2b_2} t_2^2} \nabla u(\mathbf{t}) \quad (10)$$

Consider a real function $f \in L^2(\mathbb{R}^2)$, let the boundary condition be $v_3(t_1, t_2, 0) = f(t_1, t_2)$.

Let $L_A\{v_k\}(\omega_1, \omega_2, t_3)$ be the 2D LCTs of $v_k(\mathbf{t})$, $k = 1, 2, 3$. The key to deriving GRT is to establish the relationship between $L_A\{v_1\}$, $L_A\{v_2\}$, and $L_A\{v_3\}$, as $v_3 \rightarrow f$ when $t_3 \rightarrow 0^-$. To establish the relationship between them, we first require the following crucial theorem.

THEOREM 1. *Let $L_A\{u\}(\omega_1, \omega_2, t_3)$ denote the 2D LCT of the harmonic function $u(\mathbf{t})$ with respect to t_1 and t_2 . There is*

$$L_A\{u\}(\omega_1, \omega_2, t_3) = C(\omega_1, \omega_2) e^{\sqrt{\frac{\omega_1^2}{b_1^2} + \frac{\omega_2^2}{b_2^2}} t_3} \quad (11)$$

where $t_3 < 0$, and $C(\omega_1, \omega_2)$ is independent of t_3 .

The proof of this theorem can be found in Appendix A. Based on the conclusion of this theorem, we can establish the relationship between $L_A\{v_1\}$, $L_A\{v_2\}$ and $L_A\{v_3\}$ (Theorem 2).

THEOREM 2. *Let $L_A\{v_k\}(\omega_1, \omega_2, t_3)$ be the 2D LCTs of $v_k(\mathbf{t})$, $k = 1, 2, 3$. There exists a relationship between them as follows*

$$L_A\{v_1\}(\omega_1, \omega_2, t_3) = i \frac{\omega_1/b_1}{\sqrt{(\omega_1/b_1)^2 + (\omega_2/b_2)^2}} \times L_A\{v_3\}(\omega_1, \omega_2, t_3) \quad (12)$$

$$L_A\{v_2\}(\omega_1, \omega_2, t_3) = i \frac{\omega_2/b_2}{\sqrt{(\omega_1/b_1)^2 + (\omega_2/b_2)^2}} \times L_A\{v_3\}(\omega_1, \omega_2, t_3). \quad (13)$$

Letting

$$R_A\{f\}(\omega) = (R_{A,i,1}\{f\}(\omega), R_{A,i,2}\{f\}(\omega))^\top \\ = (L_A\{v_1\}(\omega_1, \omega_2, 0), L_A\{v_2\}(\omega_1, \omega_2, 0))^\top \quad (14)$$

it follows that the relationship (Theorem 2) also holds for the continuous extension of $\mathbf{v}$ for $t_3 \rightarrow 0^-$. This $R_A\{f\}(\omega)$ represents the GRT in the linear canonical domain (Definition 1). Denoting $\omega/\mathbf{b} = (\omega_1/b_1, \omega_2/b_2)$, so that $\sqrt{(\omega_1/b_1)^2 + (\omega_2/b_2)^2} = |\omega/\mathbf{b}|$.



Figure 1: Generalized Riesz Filter

DEFINITION 1. Let $L_A\{f\}$ be the 2D LCT of signal f . Then the GRT of f is defined as

$$R_A\{f\}(\omega) = i \frac{\omega/b}{|\omega/b|} L_A\{f\}(\omega) \quad (15)$$

where $\omega = (\omega_1, \omega_2)$, $b = (b_1, b_2)$.

3.2 Generalized Riesz Filter

In the following we use the notation $\mathbf{t} = (t_1, t_2)^\top$ since $t_3 = 0$. The multiplication in the linear canonical domain in (15) corresponds to the following convolution in the spatial domain. The detailed proof process can be found in Appendix B.

THEOREM 3. The GRT of f in the spatial domain is given by

$$f_{RA}(\mathbf{t}) = e^{-i\frac{a_1}{2b_1}t_1^2} e^{-i\frac{a_2}{2b_2}t_2^2} \left[f(\mathbf{t}) e^{i\frac{a_1}{2b_1}t_1^2} e^{i\frac{a_2}{2b_2}t_2^2} * \frac{\mathbf{t}}{2\pi|\mathbf{t}|^3} \right] \\ = \frac{1}{2\pi} \text{p.v.} \iint \frac{f(\boldsymbol{\eta}) e^{i\frac{a_1}{2b_1}(t_1^2 - \eta_1^2)} e^{i\frac{a_2}{2b_2}(t_2^2 - \eta_2^2)} (\mathbf{t} - \boldsymbol{\eta})}{|\mathbf{t} - \boldsymbol{\eta}|^3} d\eta_1 d\eta_2 \quad (16)$$

where $*$ denotes 2D convolution, and $\boldsymbol{\eta} = (\eta_1, \eta_2)^\top$.

Based on the conclusion of Theorem 3, we can determine that computing the GRT of a signal f is achieved by performing the following steps: multiplying the signal by a chirp function, $e^{i\frac{a_1}{2b_1}t_1^2} e^{i\frac{a_2}{2b_2}t_2^2}$, applying a standard Riesz filter to the resulting product, and finally multiplying the output by the chirp function $e^{-i\frac{a_1}{2b_1}t_1^2} e^{-i\frac{a_2}{2b_2}t_2^2}$ (as shown in Figure 1).

4 ENVELOPE DETECTION BASED ON GRT

4.1 Generalized monogenic signal

In this section, we define the generalized monogenic signal (GMS) using the GRT.

DEFINITION 2. Let $R_{A,k}(\omega)$, $k = 1, 2$ be the GRTs of f , the GMS in linear canonical domain is defined as

$$M_A(\omega) = L_A\{f\}(\omega) - iR_{A,i,1}(\omega) - jR_{A,j,2}(\omega) \quad (17)$$

where L_A represents the 2D LCT. Its spatial-domain form is equivalent to

$$\varphi_{M_A}(\mathbf{t}) = f(\mathbf{t}) - i f_{RA,i,1}(\mathbf{t}) - j f_{RA,j,2}(\mathbf{t}). \quad (18)$$

Next, we focus on the quaternionic spectrum of the GMS, and the proof process can be found in Appendix C.

THEOREM 4. The quaternionic spectrum of the GMS is

$$L_{q,A}\{\varphi_{M_A}\}(\omega) = \left[1 - \frac{\omega_1/b_1 + \omega_2/b_2}{|\omega/b|} \right] L_A\{f\}(\omega) \quad (19)$$

where $L_{q,A}$ is the QLCT.



Figure 2: The GMS $\varphi_{G,M}(\mathbf{t})$ of Lena image with $A_1 = (0.05, 155; 1/155, 0)$, $A_2 = (0.05, 155; 1/155, 0)$. (a) $\|f_{RA}(\mathbf{t})\|$ of $\varphi_{G,M}$. (b) The phase of $\varphi_{G,M}$.

Similar to the polar decomposition of analytical signals, the polar radius (or local amplitude) of GMS is

$$\|\varphi_{M_A}(\mathbf{t})\| = \sqrt{f^2(\mathbf{t}) + \|f_{RA}(\mathbf{t})\|^2} \quad (20)$$

where $\|f_{RA}(\mathbf{t})\| = \sqrt{f_{RA,i,1}^2(\mathbf{t}) + f_{RA,j,2}^2(\mathbf{t})}$ represents the pointwise L_2 norm of the function $f_{RA}(\mathbf{t})$. The phase of GMS is

$$P(\mathbf{t}) = \arctan(\|f_{RA}(\mathbf{t})\|, f(\mathbf{t})), \mathbf{t} = (t_1, t_2). \quad (21)$$

This article utilizes the widely recognized Lena image as a common choice for testing in the field of digital image processing. The Lena image serves as a standard test image, widely employed to evaluate and benchmark digital image processing algorithms and tools. Figure 2(a) illustrates $\|f_{RA}(\mathbf{t})\|$ of the GMS for the Lena image, while Figure 2(b) represents the phase of the GMS. It is evident that they effectively enhance the edge features of the image. To further enhance the effectiveness of envelope detection, we propose a detection algorithm based on phase consistency in the subsequent section.

4.2 Envelope Detection Algorithm Based on GRT and Phase Consistency

In Section 2.3, we introduced the fundamental principle of phase consistency. In this section, we integrate this concept into the GRT space to achieve envelope detection based on GRT and phase consistency. Similar to the Fourier series, the linear canonical series of the signal f is given by [21]

$$f(t) = \sum_{n=-\infty}^{\infty} C_{A,n} \sqrt{\frac{i}{T}} \exp\left(-\frac{id}{2b} \left(n \frac{2\pi b}{T}\right)^2 + \frac{it}{b} \left(n \frac{2\pi b}{T}\right) - \frac{ia}{2b} t^2\right) \quad (22)$$

where $C_{A,n} = \int_{-T/2}^{T/2} f(t) \sqrt{\frac{-i}{T}} \exp\left(\frac{id}{2b} \left(n \frac{2\pi b}{T}\right)^2 - \frac{it}{b} \left(n \frac{2\pi b}{T}\right) + \frac{ia}{2b} t^2\right) dt$ is the expansion coefficient of a linear canonical series.

Incorporate a one-dimensional Log Gabor transform, represented by $G(\omega) = e^{\frac{-(\log(\omega/\omega_0))^2}{2(\log(\sigma/\omega_0))^2}}$ [5, 26], as a scale variable within the GRT

space

$$\mathcal{H}_{G,RA,I_k,k} = \mathbf{I}_k \frac{\omega_k/b_k}{|\omega/b|} G(|\omega/b|) \quad (23)$$

where $\mathbf{I}_k = (\mathbf{i}, \mathbf{j})$, $k = 1, 2$. Construct the GRT space by employing a ternary variable representation in the image domain

$$f_G(\mathbf{t}) = (f * G_f)(\mathbf{t}) \quad (24)$$

$$f_{G,RA,I_k,k}(\mathbf{t}) = (f * h_k)(\mathbf{t}), k = 1, 2 \quad (25)$$

where G_f is a Log Gabor transform in the image domain, h_k is the product of $\mathcal{H}_{G,RA,I_k,k}$ and the convolutional template $h(\mathbf{t}) = \frac{\mathbf{t}}{2\pi|\mathbf{t}|^3}$. Thus, the local amplitude value $A(\mathbf{t})$, local phase $P(\mathbf{t})$, and local direction of the Monological Signal after incorporating scale variables $\theta(\mathbf{t})$ can be expressed as follows

$$A(\mathbf{t}) = \sqrt{f_G^2(\mathbf{t}) + \|f_{G,RA}(\mathbf{t})\|^2} \quad (26)$$

$$P(\mathbf{t}) = \arctan(\|f_{G,RA}(\mathbf{t})\|, f_G(\mathbf{t})) \quad (27)$$

$$\theta(\mathbf{t}) = \arctan(f_{G,RA,j,2}(\mathbf{t}), f_{G,RA,i,1}(\mathbf{t})) \quad (28)$$

where $\|f_{G,RA}(\mathbf{t})\| = \sqrt{f_{G,RA,i,1}^2(\mathbf{t}) + f_{G,RA,j,2}^2(\mathbf{t})}$.

By configuring the Log Gabor transform with various wavelength parameters, which serve as the scaling factor of GRT, distinct orthogonal filters are obtained. Subsequently, the phase consistency of the image is computed. The formula for calculating the phase consistency between two factors is defined as follows

$$PC = \frac{\cos(\Delta P)}{|\sin(\Delta P) + 1|} = \frac{p_1 \cdot p_2}{|p_1 \times p_2| + |p_1| |p_2|} \quad (29)$$

$$p_k = C_{A,n} \cos(P_k), k = 1, 2 \quad (30)$$

where P_k represents the phase at the k th factor.

The algorithm process is illustrated in the Figure 3 and can be primarily divided into the following steps:

- Compute the 2D LCT of the image f to obtain $L_A\{f\}(\omega)$.
- Apply 1D Log Gabor transformations separately to the rows and columns of $L_A f(\omega)$.
- Compute $f_G(\mathbf{t})$ and $f_{G,RA,I_k,k}(\mathbf{t})$ using formulas (24) and (25) respectively. Then, obtain the first set of vectors p_1 for calculating phase consistency.
- By varying the wavelength of the 1D Log Gabor transform, a second set of vectors p_2 is obtained for calculating phase consistency.
- It is also feasible to obtain multiple sets of vectors p_k , calculate phase consistency results multiple times, and subsequently perform image fusion on each result. This approach aims to achieve improved extraction of low-level features.
- Extracting Image feature edges with non-maximum suppression.

We have chosen five widely used operators, namely Sobel, Roberts, Prewitt, Canny, and Zero-cross, for comparison, as shown in Figure 4. By considering the example where $A_1 = A_2 = (0.05, 155; 1/155, 0)$, it becomes apparent that our method demonstrates outstanding performance in areas with unclear edges.

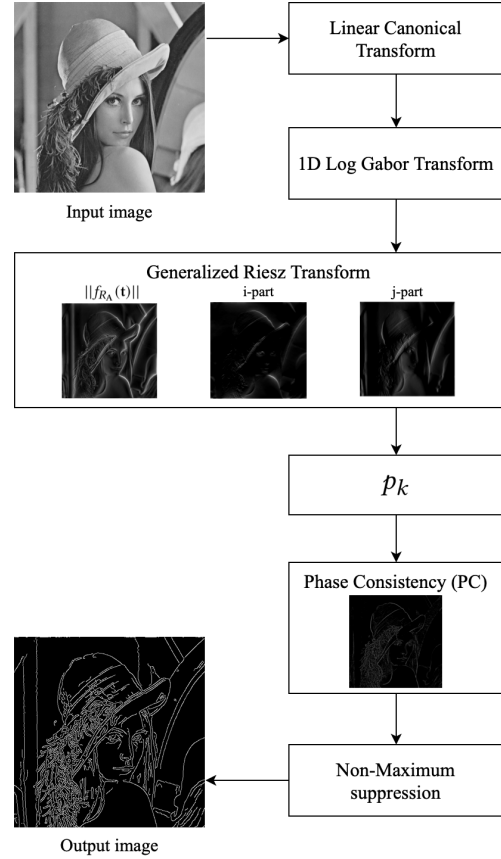


Figure 3: Algorithm flow chart.

5 CONCLUSION

In conclusion, this article presents an effective spatial domain filtering formula to address negative frequency components by defining the Generalized Riesz Transform (GRT) in the linear canonical domain. The GRT is constructed using a three-dimensional harmonic field and leverages the properties of the linear canonical transform. The effectiveness of the GRT in capturing texture features is demonstrated through the introduction of the concept of the generalized monogenic signal. Furthermore, an envelope detection algorithm based on GRT and phase consistency is proposed, showcasing superior performance compared to conventional methods. Overall, the GRT offers a promising approach to enhance feature extraction capabilities in various fields, contributing to the advancement of signal and image processing technologies.

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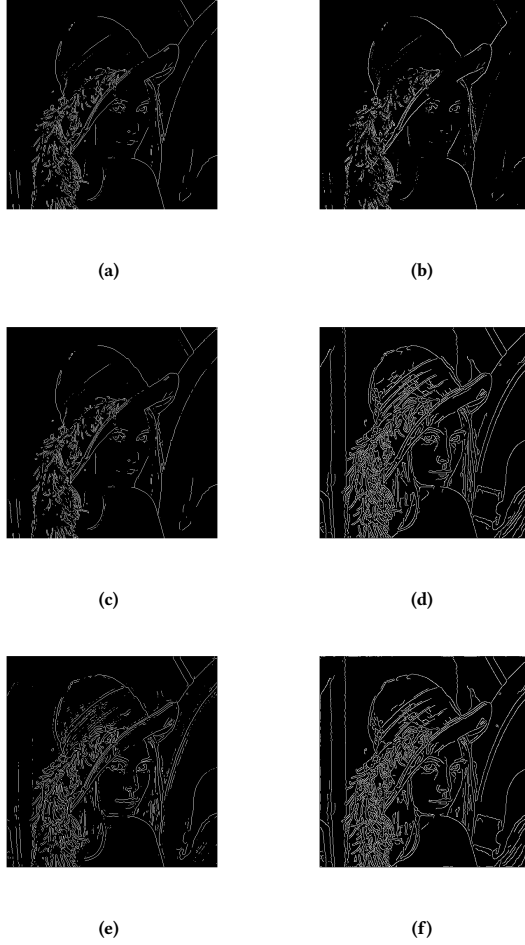


Figure 4: Comparison of envelope detection effects using different methods (Lena image). (a) Sobel operator. (b) Roberts operator. (c) Prewitt operator. (d) Canny operator. (e) Zero-cross method. (f) Proposed method.

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A PROOF OF THEOREM 1

The harmonic potential field of $u(t)$ is irrotational and solenoidal in the open domain $t_3 < 0$ [4]. Then the following equation holds

$$\begin{aligned} \operatorname{div}(\nabla u(t)) &= \nabla \cdot \nabla u(t) = \frac{\partial}{\partial t_1} e^{i \frac{a_1}{2b_1} t_1^2} e^{i \frac{a_2}{2b_2} t_2^2} v_1 \\ &+ \frac{\partial}{\partial t_2} e^{i \frac{a_1}{2b_1} t_1^2} e^{i \frac{a_2}{2b_2} t_2^2} v_2 + \frac{\partial}{\partial t_3} e^{i \frac{a_1}{2b_1} t_1^2} e^{i \frac{a_2}{2b_2} t_2^2} v_3 = 0 \end{aligned} \quad (31)$$

where $\operatorname{div}(\cdot)$ represents divergence.

Apply the 2D LCT to (31) in the t_1 and t_2 directions with $A = (A_1, A_2)$, where $A_k = (a_k, b_k; c_k, d_k)$, $a_k d_k - b_k c_k = 1$, and $b_k \neq 0$ for $k = 1, 2$. Then

$$\begin{aligned} L_A\{\operatorname{div}(\nabla u(t))\} &= L_A\{e^{-i \frac{a_1}{2b_1} t_1^2} e^{-i \frac{a_2}{2b_2} t_2^2} \frac{\partial}{\partial t_1} e^{i \frac{a_1}{2b_1} t_1^2} e^{i \frac{a_2}{2b_2} t_2^2} v_1\} \\ &+ L_A\{e^{-i \frac{a_1}{2b_1} t_1^2} e^{-i \frac{a_2}{2b_2} t_2^2} \frac{\partial}{\partial t_2} e^{i \frac{a_1}{2b_1} t_1^2} e^{i \frac{a_2}{2b_2} t_2^2} v_2\} + \frac{\partial}{\partial t_3} L_A\{v_3\} = 0. \end{aligned} \quad (32)$$

By utilizing the differential properties of the 2D FT, the first term and the second term on the left side of (32) can be derived as follows

$$\begin{aligned}
& L_A \{ e^{-i\frac{a_1}{2b_1}t_1^2} e^{-i\frac{a_2}{2b_2}t_2^2} \frac{\partial}{\partial t_k} e^{i\frac{a_1}{2b_1}t_1^2} e^{i\frac{a_2}{2b_2}t_2^2} v_k \} (\omega_1, \omega_2, t_3) \\
&= \frac{1}{\sqrt{i2\pi b_1}} \frac{1}{\sqrt{i2\pi b_2}} e^{i(\frac{d_1}{2b_1}\omega_1^2 + \frac{d_2}{2b_2}\omega_2^2)} i \frac{\omega_k}{b_k} \iint e^{i\frac{a_1}{2b_1}t_1^2} e^{i\frac{a_2}{2b_2}t_2^2} v_k(t) \\
&\quad \times e^{-i(\frac{1}{b_1}t_1\omega_1 + \frac{1}{b_2}t_2\omega_2)} dt_1 dt_2 \\
&= i \frac{\omega_k}{b_k} L_A \{ v_k \} (\omega_1, \omega_2, t_3)
\end{aligned} \tag{33}$$

Taking (33) into (32), there is

$$i \frac{\omega_1}{b_1} L_A \{ v_1 \} + i \frac{\omega_2}{b_2} L_A \{ v_2 \} + \frac{\partial}{\partial t_3} L_A \{ v_3 \} = 0. \tag{34}$$

Since $v_k(t) = e^{-i\frac{a_1}{2b_1}t_1^2} e^{-i\frac{a_2}{2b_2}t_2^2} \frac{\partial}{\partial t_k} u(t)$, $k = 1, 2, 3$, the equation (34) can be rewritten as

$$i^2 \frac{\omega_1^2}{b_1^2} L_A \{ u \} + i^2 \frac{\omega_2^2}{b_2^2} L_A \{ u \} + \frac{\partial^2}{\partial t_3^2} L_A \{ u \} = 0.$$

Furthermore, the above equation can be simplified as

$$- \left(\sqrt{\frac{\omega_1^2}{b_1^2} + \frac{\omega_2^2}{b_2^2}} \right)^2 L_A \{ u \} (\omega_1, \omega_2, t_3) + \frac{\partial^2}{\partial t_3^2} L_A \{ u \} (\omega_1, \omega_2, t_3) = 0.$$

This differential equation is solved for $t_3 < 0$ by

$$L_A \{ u \} (\omega_1, \omega_2, t_3) = C(\omega_1, \omega_2) e^{\sqrt{\frac{\omega_1^2}{b_1^2} + \frac{\omega_2^2}{b_2^2}} t_3}$$

where $C(\omega_1, \omega_2)$ is independent of t_3 .

B PROOF OF THEOREM 3

Let $f_{R_A}(t) = (f_{R_{A,i,1}}(t), f_{R_{A,i,2}}(t))^T$ be the GRT of f in the spatial domain, where $f_{R_{A,i,k}}(t) = L_A^{-1} \{ R_{A,i,k}(\omega) \}$, $k = 1, 2$. Let's begin by considering the first term $f_{R_{A,i,1}}(t)$ in $f_{R_A}(t)$:

$$\begin{aligned}
f_{R_{A,i,1}}(t) &= L_A^{-1} \{ R_{A,i,1}(\omega) \} = \frac{1}{\sqrt{-i2\pi b_1}} \frac{1}{\sqrt{-i2\pi b_2}} \\
&\iint \frac{i\omega_1/b_1 L_A \{ f \} (\omega_1, \omega_2)}{\sqrt{(\omega_1/b_1)^2 + (\omega_2/b_2)^2}} e^{-i(\frac{a_1}{2b_1}\eta_1^2 - \frac{1}{b_1}\eta_1\omega_1 + \frac{a_1}{2b_1}\omega_1^2)} \\
&\times e^{-i(\frac{a_2}{2b_2}\eta_2^2 - \frac{1}{b_2}\eta_2\omega_2 + \frac{a_2}{2b_2}\omega_2^2)} d\omega_1 d\omega_2 \\
&= \frac{-e^{i\frac{a_1}{2b_1}\eta_1^2} e^{-i\frac{a_2}{2b_2}\eta_2^2}}{4\pi^2 |b_1| |b_2|} \iint F \{ f(t_1, t_2) e^{i\frac{a_1}{2b_1}t_1^2} e^{i\frac{a_2}{2b_2}t_2^2} \} (\omega_1, \omega_2) \\
&\times \frac{i\omega_1/b_1}{\sqrt{(\omega_1/b_1)^2 + (\omega_2/b_2)^2}} e^{i(\frac{1}{b_1}\eta_1\omega_1 + \frac{1}{b_2}\eta_2\omega_2)} d\omega_1 d\omega_2.
\end{aligned}$$

By taking the convolution theorem of FT in frequency domain, it follows that

$$\begin{aligned}
f_{R_{A,i,1}}(t) &= e^{-i\frac{a_1}{2b_1}t_1^2} e^{-i\frac{a_2}{2b_2}t_2^2} [f(t_1, t_2) e^{i\frac{a_1}{2b_1}t_1^2} e^{i\frac{a_2}{2b_2}t_2^2} \\
&\quad * F^{-1} \left\{ \frac{i\omega_1/b_1}{\sqrt{(\omega_1/b_1)^2 + (\omega_2/b_2)^2}} \right\} (t_1, t_2)]
\end{aligned}$$

where F^{-1} represents the inverse 2D FT. Then

$$\begin{aligned}
F^{-1} \left\{ \frac{i\omega_1/b_1}{\sqrt{(\omega_1/b_1)^2 + (\omega_2/b_2)^2}} \right\} (t_1, t_2) &= \frac{1}{4\pi^2} \\
&\times \iint \frac{i\omega_1/b_1 e^{i(\frac{1}{b_1}t_1\omega_1 + \frac{1}{b_2}t_2\omega_2)}}{\sqrt{(\omega_1/b_1)^2 + (\omega_2/b_2)^2}} d\frac{\omega_1}{b_1} d\frac{\omega_2}{b_2} = -\frac{1}{\pi^2} \int_0^\infty \left(\cos \frac{\omega_2}{b_2} t_2 \right) d\frac{\omega_2}{b_2} \\
&\times \int_0^\infty \frac{\omega_1/b_1}{\sqrt{(\omega_1/b_1)^2 + (\omega_2/b_2)^2}} \left(\sin \frac{\omega_1}{b_1} t_1 \right) d\frac{\omega_1}{b_1}.
\end{aligned} \tag{35}$$

Applying the integral formula 3.773.3 from [6], there is

$$\begin{aligned}
\int_0^\infty \frac{\omega_1/b_1 \left(\sin \frac{\omega_1}{b_1} t_1 \right)}{\sqrt{(\omega_1/b_1)^2 + (\omega_2/b_2)^2}} d\frac{\omega_1}{b_1} &= -\frac{\sqrt{\pi}}{\Gamma(\frac{1}{2})} \frac{d}{dt_1} \left[B_0 \left(\frac{\omega_2}{b_2} t_2 \right) \right] \\
&= \frac{\omega_2}{b_2} B_1 \left(\frac{\omega_2}{b_2} t_2 \right)
\end{aligned}$$

where $\Gamma(\cdot)$ is the Gamma function and $B_n(\cdot)$ are the modified Bessel functions. The equation (35) can be rewritten as

$$\begin{aligned}
F^{-1} \left\{ \frac{i\omega_1/b_1}{\sqrt{(\omega_1/b_1)^2 + (\omega_2/b_2)^2}} \right\} (t_1, t_2) \\
= -\frac{1}{\pi^2} \int_0^\infty \frac{\omega_2}{b_2} K_1 \left(\frac{\omega_2}{b_2} t_2 \right) \left(\cos \frac{\omega_2}{b_2} t_2 \right) d\frac{\omega_2}{b_2}.
\end{aligned}$$

By using the expression 6.669.12 from [6], it follows that

$$\begin{aligned}
F^{-1} \left\{ \frac{i\omega_1/b_1}{\sqrt{(\omega_1/b_1)^2 + (\omega_2/b_2)^2}} \right\} (t_1, t_2) \\
= -\frac{1}{2\pi^2} (2t_1) \Gamma \left(\frac{3}{2} \right) \left(t_1^2 + t_2^2 \right)^{-\frac{3}{2}} = \frac{1}{2\pi} \frac{t_1}{(t_1^2 + t_2^2)^{3/2}}.
\end{aligned}$$

Therefore, the GRT in the spatial domain is

$$\begin{aligned}
f_{R_{A,i,1}}(t) &= e^{-i\frac{a_1}{2b_1}t_1^2} e^{-i\frac{a_2}{2b_2}t_2^2} \left[f(t_1, t_2) e^{i\frac{a_1}{2b_1}t_1^2} e^{i\frac{a_2}{2b_2}t_2^2} * \frac{1}{2\pi} \frac{t_1}{(t_1^2 + t_2^2)^{3/2}} \right] \\
&= \frac{1}{2\pi} \text{p.v.} \iint \frac{f(t_1, t_2) e^{i\frac{a_1}{2b_1}(t_1^2 - \eta_1^2)} e^{i\frac{a_2}{2b_2}(t_2^2 - \eta_2^2)} (t_1 - \eta_1)}{[(t_1 - \eta_1)^2 + (t_2 - \eta_2)^2]^{3/2}} d\eta_1 d\eta_2.
\end{aligned}$$

In a similar fashion, we can obtain the form of $f_{R_{A,i,2}}(t)$. The final result is obtained by expressing the above results in vector form.

C PROOF OF THEOREM 4

A real 2-D signal can be written in the form $f = f_{ee} + f_{oe} + f_{eo} + f_{oo}$, where f_{ee} denotes the part of f which is even with respect to t_1 and t_2 , f_{oe} denotes the part which is odd with respect to t_1 and even with respect to t_2 and so on. The QLCT of f has the form

$$\begin{aligned}
L_{q,A} \{ f \} (\omega) &= \iint \frac{e^{-i(\frac{\theta_1}{2} + \alpha_1)}}{\sqrt{r_1}} (f_{ee} + f_{oe} + f_{eo} + f_{oo}) \frac{e^{-j(\frac{\theta_2}{2} + \alpha_2)}}{\sqrt{r_2}} dt_1 dt_2 \\
&= L_{ee,q,A} \{ f \} (\omega) - i L_{oe,q,A} \{ f \} (\omega) - j L_{eo,q,A} \{ f \} (\omega) + k L_{oo,q,A} \{ f \} (\omega)
\end{aligned}$$

where $\alpha_k = -\left(\frac{a_k}{2b_k} \tau_k^2 - \frac{1}{b_k} \tau_k \omega_k + \frac{d_k}{2b_k} \omega_k^2 \right)$, $r_k = \sqrt{4\pi^2 b_k^2}$, $\theta_k = \arctan(2\pi b_k)$, $k = 1, 2$.

The quaternionic spectrum of $f_{R_{p,i,1}}(t)$ is $L_{q,A} \{ f_{R_{A,i,1}} \} (\omega) = R_{A,i,1} \{ f \} (\omega) = [i L_{ee,q,A} \{ f \} (\omega) + L_{oe,q,A} \{ f \} (\omega) - k L_{eo,q,A} \{ f \} (\omega) - j L_{oo,q,A} \{ f \} (\omega)] \frac{\omega_1/b_1}{|\omega/b|}$.

Similarly, $L_{q,A}\{f_{R_{A,j,2}}\}(\omega) = [jL_{ee,q,A}\{f\}(\omega) + kL_{oe,q,A}\{f\}(\omega) + L_{eo,q,A}\{f\}(\omega) + iL_{oo,q,A}\{f\}(\omega)] \frac{\omega_2/b_2}{|\omega/b|}$. Substituting the above equation into $L_{q,A}\{\varphi_{M_A}\}$ yields the desired result.